Linear Perturbation Theory for General Relativity: Application to Cosmology

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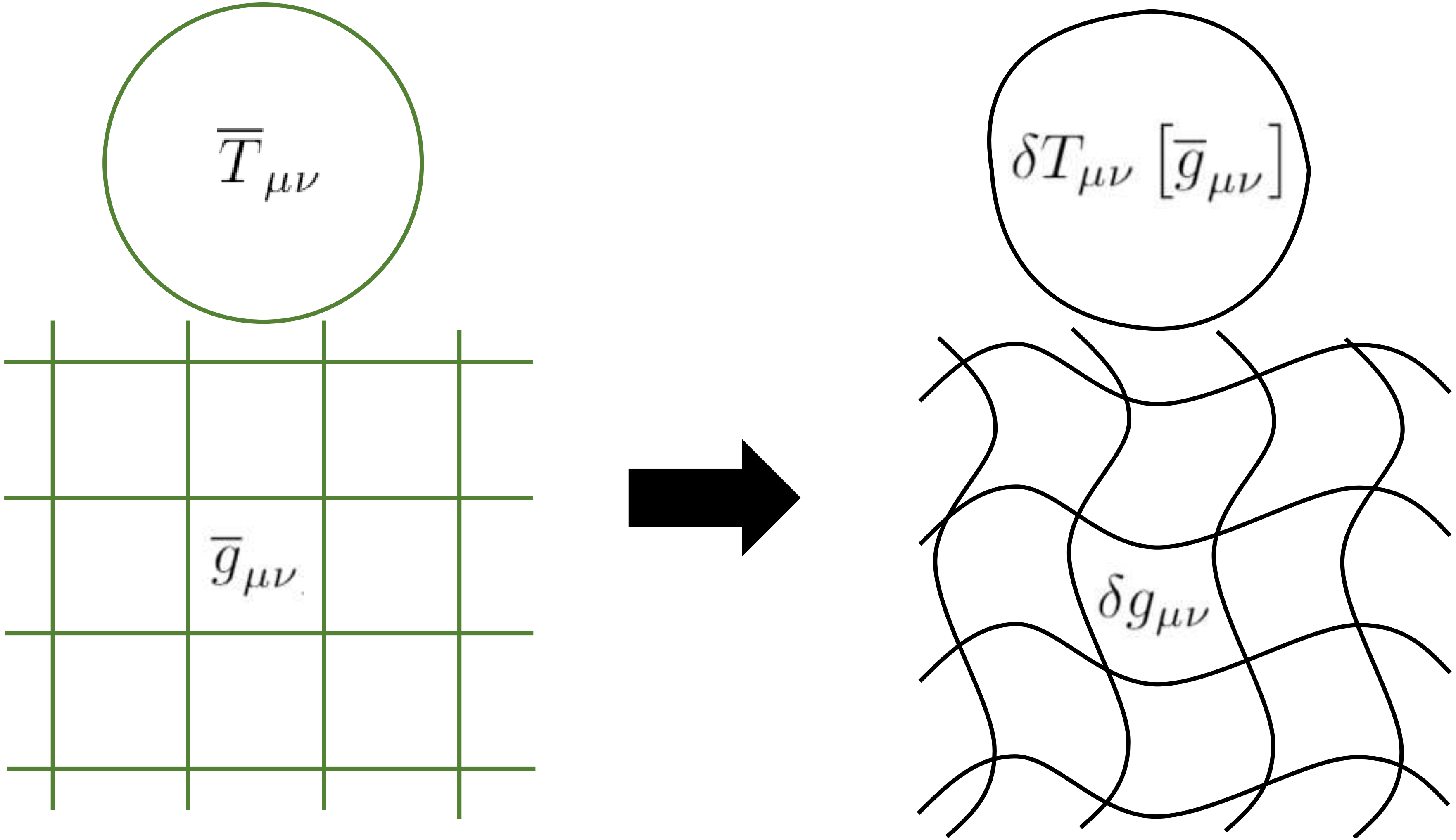
1. Perturbation theory in GR



$$\overline{G}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]=\overline{T}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right] \qquad G_{\mu\nu}\left[\overline{g}_{\mu\nu}+\delta g_{\mu\nu}\right]=\overline{T}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]+\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$$

1. Perturbation theory in GR

Given a simple $\overline{T}_{\mu\nu}$ For which we know how to compute a $\overline{g}_{\mu\nu}$ such that $\overline{G}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]=\overline{T}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$

If we now introduce a small perturbation in the stress-energy tensor which we can express using the metric of the unperturbed scenario $\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$ How can we find a (also small) perturbation of the metric $\delta g_{\mu\nu}$ such that the following equation is satisfied $G_{\mu\nu}\left[\overline{g}_{\mu\nu}+\delta g_{\mu\nu}\right]=\overline{T}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]+\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$?

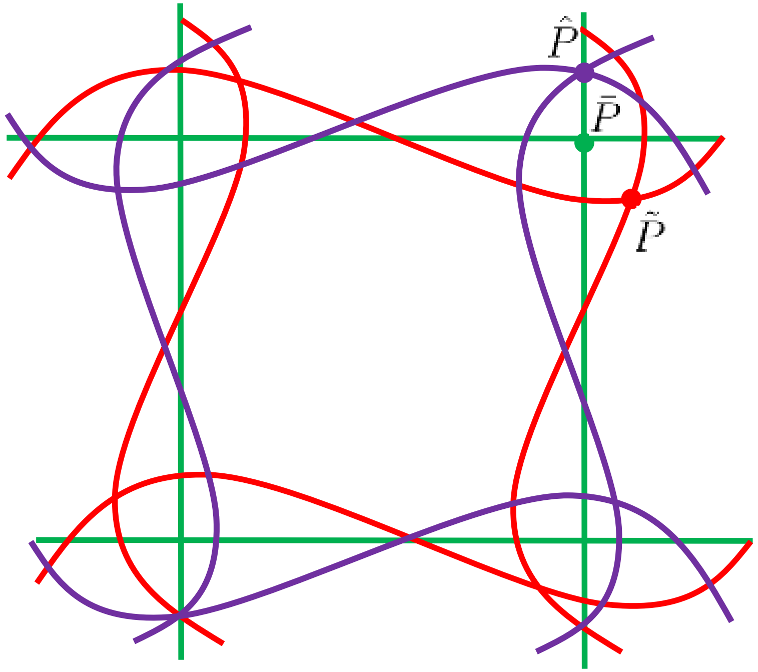
For a sufficiently small $\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$ we have that $\delta g_{\mu\nu}$ is also small enough in order to write the following expansion $G_{\mu\nu}\left[\overline{g}_{\mu\nu}+\delta g_{\mu\nu}\right]=\overline{G}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]+\delta G_{\mu\nu}\left[\overline{g}_{\mu\nu},\delta g_{\mu\nu}\right]+\cancelto{0}{\mathcal{O}(\delta g^2)}$ Hence:

$$\overline{G}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]+\delta G_{\mu\nu}\left[\overline{g}_{\mu\nu},\delta g_{\mu\nu}\right]=\overline{T}_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]+\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]\longrightarrow\delta G_{\mu\nu}\left[\overline{g}_{\mu\nu},\delta g_{\mu\nu}\right]=\delta T_{\mu\nu}\left[\overline{g}_{\mu\nu}\right]$$

2. Gauge freedom

Problem: We use a particular chart to explicitly express the metric and other quantities. Depending on the chart chosen, the expression for these quantities will change. We are only interested in the objects that we can construct which are invariant under coordinate transformations.

To analyze how different fields behave under coordinate transformations we take three different points that have the same coordinates when expressed in different coordinate systems, that is, the point $\bar{P}$ with coordinates $\bar{x}^\mu(\bar{P})$ in a coordinate system defined over the unperturbed scenario, the point $\hat{P}$ with the same coordinates $\hat{x}^\mu(\hat{P})$ when expressed in a particular system defined in the perturbed spacetime, and the point $\tilde{P}$ with also the same coordinates $\tilde{x}^\mu(\tilde{P})$ when expressed in another system defined over the perturbed spacetime: $\bar{x}^\mu(\bar{P}) = \hat{x}^\mu(\hat{P}) = \tilde{x}^\mu(\tilde{P})$



Imagine a vector field ξ^μ that allows us to obtain the coordinates of a point expressed in one of the perturbed coordinate systems knowing the coordinates of the same point expressed in the other perturbed coordinate system as:

$$\tilde{x}^\mu(P) = \hat{x}^\mu(P) + \xi^\mu(P)$$

Then for the points $\hat{P}$ and $\tilde{P}$:

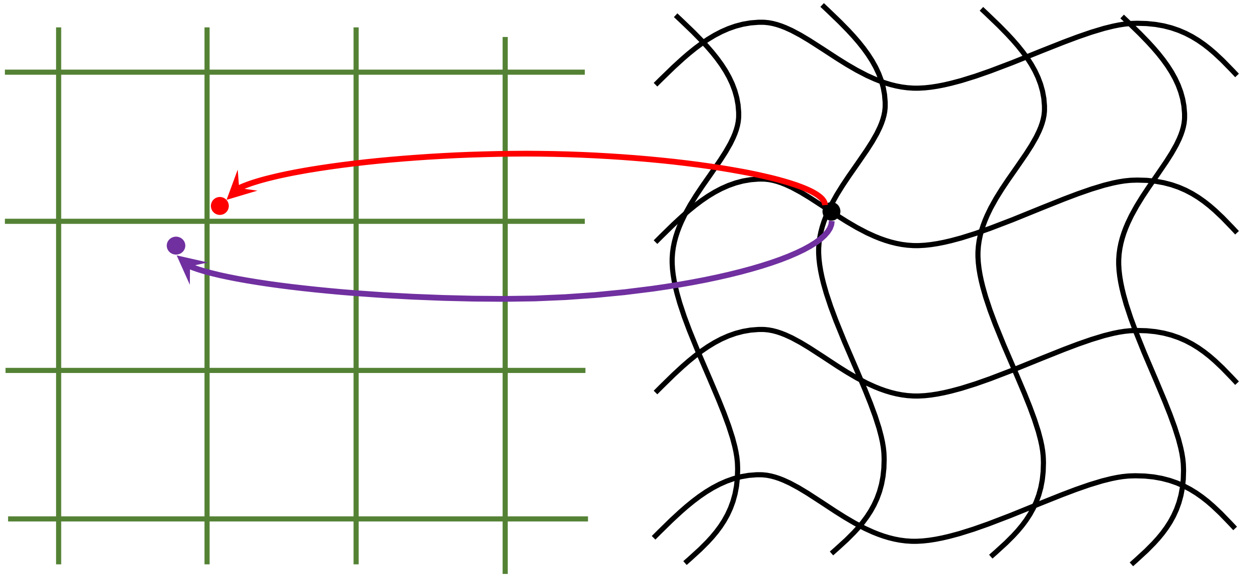
$$\left\{ \begin{array}{l} \tilde{x}^\mu(\tilde{P}) = \hat{x}^\mu(\tilde{P}) + \xi^\mu(\tilde{P}) \\ \tilde{x}^\mu(\hat{P}) = \hat{x}^\mu(\hat{P}) + \xi^\mu(\hat{P}) \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \hat{x}^\mu(\tilde{P}) = \hat{x}^\mu(\hat{P}) - \xi^\mu(\tilde{P}) \\ \tilde{x}^\mu(\tilde{P}) = \tilde{x}^\mu(\hat{P}) - \xi^\mu(\hat{P}) \end{array} \right.$$

2. Gauge freedom

Now imagine a scalar field originally defined over the unperturbed spacetime. The value of this field at a particular point in the background would be $\bar{s}(\bar{P})$. The point associated to $\bar{P}$ in the perturbed tilde coordinate system is $\tilde{P}$, and the value of the perturbed field here is $s(\tilde{P})$, however the point associated to $\bar{P}$ in the perturbed hatted coordinate system is $\hat{P}$, and the value of the perturbed field here is $s(\hat{P})$. The value of the perturbed field at a particular point should not depend upon the coordinate system we choose to describe the spacetime, that is:

$$s(\hat{P}) = s(\hat{x}^\mu(\hat{P})) = s(\tilde{x}^\mu(\hat{P})) = s(\tilde{x}^\mu(\tilde{P}) + \xi^\mu(\hat{P}))$$

The same point in the perturbed system is associated with different points in the background spacetime depending on the coordinate system we choose to use, therefore the perturbation of a field (which we define over the points of the unperturbed spacetime) will depend on the coordinate system chosen



2. Gauge freedom

We can mathematically express the difference between the perturbations written in different coordinate systems as follows:

$$\underbrace{\begin{gathered} s(\tilde{P}) - \bar{s}(\bar{P}) = \tilde{\delta} s(\bar{P}) \\ s(\hat{P}) - \bar{s}(\bar{P}) = \hat{\delta} s(\bar{P}) \end{gathered}}$$

$$\tilde{\delta} s(\bar{P}) - \hat{\delta} s(\bar{P}) = s(\tilde{P}) - s(\hat{P})$$

$$\hat{\delta} s(\bar{P}) - \tilde{\delta} s(\bar{P}) = \xi^{\alpha}(\tilde{P}) \left. \frac{\partial s}{\partial \hat{x}^{\alpha}} \right|_{\hat{P}}$$

$$\hat{\delta} s(\bar{P}) - \tilde{\delta} s(\bar{P}) = \xi^{\alpha}(\bar{P}) \left. \frac{\partial s}{\partial \bar{x}^{\alpha}} \right|_{\bar{P}}$$

If we perform a Taylor expansion of $s(\hat{x}^{\mu})$ around $\hat{P}$:

$$s(\hat{x}^{\mu}) = s(\hat{x}^{\mu}(\hat{P})) + \left. \frac{\partial s}{\partial \hat{x}^{\alpha}} \right|_{\hat{P}} \left(\hat{x}^{\alpha} - \hat{x}^{\alpha}(\hat{P}) \right) + \cancelto{0}{\mathcal{O}^2}$$

Evaluating the former expression at $\hat{x}^{\mu}(\tilde{P})$:

$$s(\tilde{P}) \approx s(\hat{P}) + \left. \frac{\partial s}{\partial \hat{x}^{\alpha}} \right|_{\hat{P}} \left(\hat{x}^{\alpha}(\tilde{P}) - \hat{x}^{\alpha}(\hat{P}) \right) \rightarrow s(\tilde{P}) \approx s(\hat{P}) - \xi^{\alpha}(\tilde{P}) \left. \frac{\partial s}{\partial \hat{x}^{\alpha}} \right|_{\hat{P}}$$

$\hat{x}^{\mu}(\tilde{P}) = \hat{x}^{\mu}(\hat{P}) - \xi^{\mu}(\tilde{P})$ (From slide 5)

We now take into account that the difference $\xi^{\alpha}(\tilde{P}) - \xi^{\alpha}(\hat{P})$ second order small, hence we can write

$\xi^{\mu}(\tilde{P}) = \xi^{\mu}(\hat{P}) = \xi^{\mu}(\bar{P}) = \xi^{\mu}(\bar{x}^{\mu})$ at first order. Also at first order $\left. \frac{\partial s}{\partial \hat{x}^{\mu}} \right|_{\hat{P}} = \left. \frac{\partial s}{\partial \bar{x}^{\mu}} \right|_{\bar{P}}$

2. Gauge freedom

If we now follow an analogous approach for a twice covariant tensor field we have:

$$\underbrace{\begin{aligned} B_{\tilde\mu\tilde\nu}(\tilde P) - \bar B_{\mu\nu}(\bar P) &= \tilde\delta B_{\mu\nu}(\bar P) \\ B_{\hat\mu\hat\nu}(\hat P) - \bar B_{\mu\nu}(\bar P) &= \hat\delta B_{\mu\nu}(\bar P) \end{aligned}}$$

$$\tilde\delta B_{\mu\nu}(\bar P) - \hat\delta B_{\mu\nu}(\bar P) = B_{\tilde\mu\tilde\nu}(\tilde P) - B_{\hat\mu\hat\nu}(\hat P)$$

$$\hat\delta B_{\mu\nu}(\bar P) - \tilde\delta B_{\mu\nu}(\bar P) = \partial_\mu \xi^\alpha \bar B_{\alpha\nu} + \partial_\nu \xi^\alpha \bar B_{\mu\alpha} + \xi^\alpha \partial_\alpha \bar B_{\mu\nu}$$

If we perform a Taylor expansion of $B_{\hat\mu\hat\nu}(\hat x^\mu)$ around $\hat P$:

$$B_{\hat\mu\hat\nu}(\hat x^\mu) = B_{\hat\mu\hat\nu}(\hat x^\mu(\hat P)) + \left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \hat x^\alpha}\right|_{\hat P} \left(\hat x^\alpha - \hat x^\alpha(\hat P)\right) + \cancelto{0}{\mathcal{O}^2}$$

Evaluating the former expression at $\hat x^\mu(\tilde P)$:

$$B_{\hat\mu\hat\nu}(\tilde P) \approx B_{\hat\mu\hat\nu}(\hat P) + \left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \hat x^\alpha}\right|_{\hat P} \left(\hat x^\alpha(\tilde P) - \hat x^\alpha(\hat P)\right) \longrightarrow B_{\hat\mu\hat\nu}(\tilde P) \approx B_{\hat\mu\hat\nu}(\hat P) - \xi^\alpha(\tilde P)\left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \hat x^\alpha}\right|_{\hat P}$$

$\hat x^\mu(\tilde P) = \hat x^\mu(\hat P) - \xi^\mu(\tilde P)$ (From slide 5)

A twice covariant tensor transforms as: $B_{\tilde\mu\tilde\nu}(\tilde P) = \frac{\partial \hat x^\rho}{\partial \tilde x^\mu}\frac{\partial \hat x^\sigma}{\partial \tilde x^\nu} B_{\hat\mu\hat\nu}(\tilde P) \longrightarrow$ …

$\tilde x^\mu(P) = \hat x^\mu(P) + \xi^\mu(P)$ (From slide 5)

… $\longrightarrow B_{\tilde\mu\tilde\nu}(\tilde P) = \left(\delta^\rho_\mu - \partial_\mu \xi^\rho\right)\left(\delta^\sigma_\nu - \partial_\nu \xi^\sigma\right)\left[B_{\hat\mu\hat\nu}(\hat P) - \xi^\alpha(\tilde P)\left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \hat x^\alpha}\right|_{\hat P}\right]$

Taking into account as in the previous slide that at first order $\xi^\mu(\tilde P) = \xi^\mu(\hat P) = \xi^\mu(\bar P) = \xi^\mu(\bar x^\mu)$ and

$\left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \hat x^\alpha}\right|_{\hat P} = \left.\frac{\partial B_{\hat\mu\hat\nu}}{\partial \bar x^\alpha}\right|_{\bar P}$ The expression reduces at first order to: $B_{\tilde\mu\tilde\nu}(\tilde P) = B_{\hat\mu\hat\nu}(\hat P) - \partial_\mu \xi^\alpha \bar B_{\alpha\nu} - \partial_\nu \xi^\alpha \bar B_{\mu\alpha} - \xi^\alpha \partial_\alpha \bar B_{\mu\nu}$

2. Gauge freedom

Therefore particularizing the expression of the last slide for the metric tensor we have :

$$\delta\hat{g}_{\mu\nu}(\bar{P}) - \delta\tilde{g}_{\mu\nu}(\bar{P}) = \partial_\mu\xi^\alpha\bar{g}_{\alpha\nu} + \partial_\nu\xi^\alpha\bar{g}_{\mu\alpha} + \xi^\alpha\partial_\alpha\bar{g}_{\mu\nu} \rightarrow \Delta_\xi\delta g_{\mu\nu} = \mathscr{L}_\xi\bar{g}_{\mu\nu}$$

We can also apply the same procedure as presented here for a scalar field and a twice covariant tensor to other tensor fields of different rank. For example for a vector field and a (1-covariant, 1-contravariant) tensor:

$$\widetilde{\delta w}^\alpha = \widehat{\delta w}^\alpha + \xi^\alpha{}_{,\beta}\bar{w}^\beta - \bar{w}^\alpha{}_{,\beta}\xi^\beta$$

$$\widetilde{\delta A}_\nu^\mu = \widehat{\delta A}_\nu^\mu + \xi^\mu{}_{,\rho}\bar{A}_\nu^\rho - \xi^\sigma{}_{,\nu}\bar{A}_\sigma^\mu - \bar{A}_{\nu,\alpha}^\mu\xi^\alpha$$

Now we are ready to study which quantities are gauge invariant for a FLRW metric…

3. Gauge invariant quantities for the FLRW metric

The FLRW background metric can be written in terms of the conformal time ($d\eta = dt/a(t)$) as:

$$ds^2 = \bar{g}_{\mu\nu} dx^\mu dx^\nu = -a(\eta)^2 d\eta^2 + a(\eta)^2 \delta_{ij} dx^i dx^j$$

If we now give the following names to the terms of the perturbed metric:

$$[\delta g_{\mu\nu}] = a^2 \begin{bmatrix} -2A & -B_i \\ -B_i & -2D\delta_{ij} + 2E_{ij} \end{bmatrix}$$

And apply the magical formula derived in section 2 for instance for the 00 term we obtain:

$$\Delta_\xi \delta g_{00} = \mathscr{L}_\xi \bar{g}_{00} \rightarrow \Delta_\xi \left(-2a^2 A\right) = 2a^2 \partial_0 \xi^0 + 2aa'\xi^0 \rightarrow \Delta_\xi A = -\partial_0 \xi^0 - \frac{a'}{a}\xi^0$$

Following the same approach for the other components (0i, ij) and doing some algebraic tricks:

$$\Delta_\xi B_i = \partial_0 \xi^i - \partial_i \xi^0$$

$$\Delta_\xi D = \frac{1}{3}\partial_k \xi^k + \frac{a'}{a}\xi^0$$

$$\Delta_\xi E_{ij} = -\frac{1}{2}\left(\partial_j \xi^i + \partial_i \xi^j\right) + \frac{1}{3}\delta_{ij}\partial_k \xi^k$$

3. Gauge invariant quantities for the FLRW metric

SVT decomposition